

Baby Names

1. Baby name persistence

First, I get the national names counts, or, the sum of each name across all the states for each year and gender

Second, I rank all the names by their count (rank 1 being the most popular), within each year-gender pairing.

Third, I create a function for the top n ranked names for year-gender pairings

Fourth, in order to measure persistence, I tracked each each years top 25 names (male and female), and then checked where they ranked 1, 2 and 3 years later. The Spearman rank correlation is used, where a rank close to 1 means the names stayed in the same order, and close to 0 means they changed significantly.

Fifth, I plot the correlation over time.

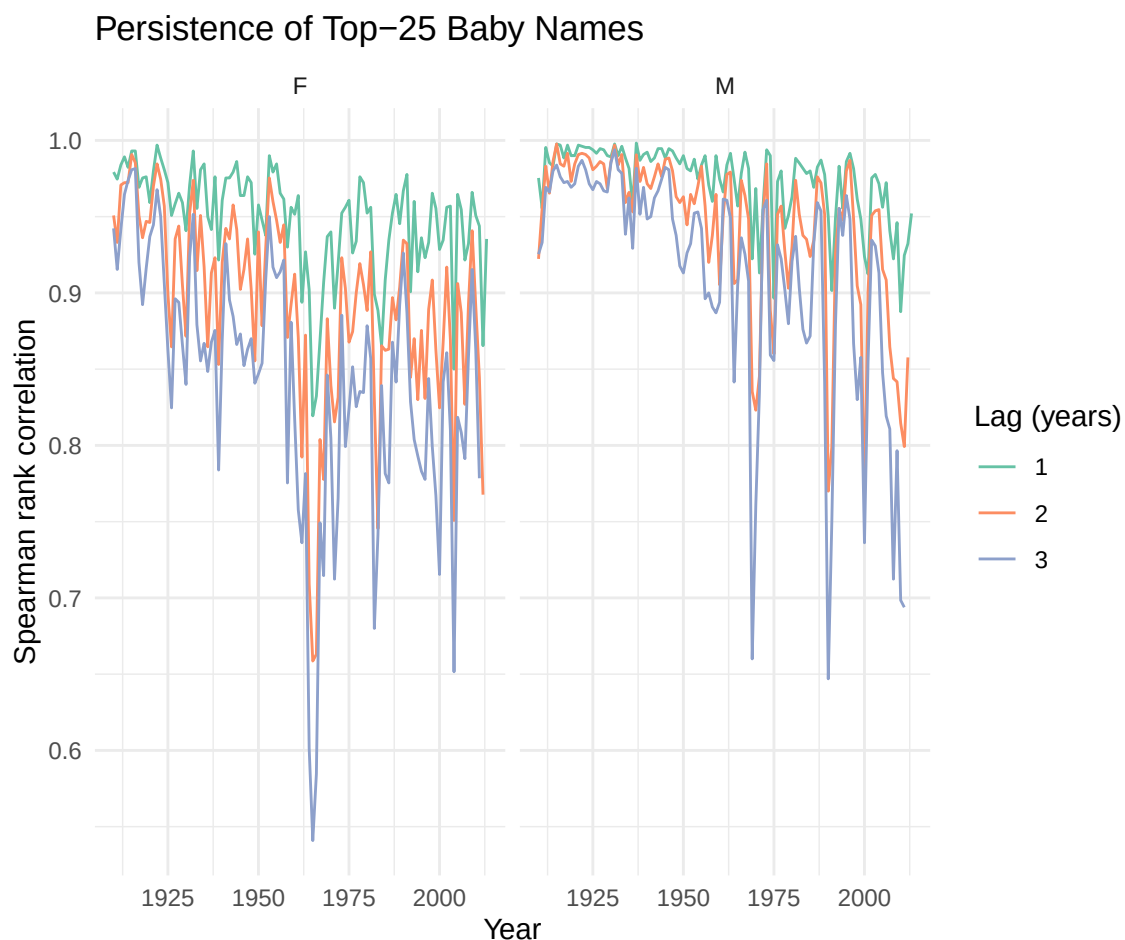


Figure 1.1: The Persistence of Baby Names.

The plot does indicate that the agencies suspicions may have been correct, as there does appear to be a drop-off in persistence of the top 25 names from around 1990. The following table further confirms the agencies suspicion. At each lag, the mean correlation drops from the pre- to post-1990 period. This is possibly attributable to cultural influences such as social media and streaming causing more frequent waves of popular names. This motivates the subsequent section, which investigates drivers of baby names.

Table 1.1: Mean Correlation around 1990

Period	Lag (years)	Mean Correlation
Post 1990	1	0.944
Pre 1990	1	0.966
Post 1990	2	0.879
Pre 1990	2	0.929

Period	Lag (years)	Mean Correlation
Post 1990	3	0.825
Pre 1990	3	0.893

2. Investigating naming spikes

First, I check the distribution of the data in order to get a threshold which will be use to filter out low-volume names. 500 is chosen as the threshold.

Then, I calculate the naming spikes as year-on-year percentage changes.

Table 2.1: Top 10 Year-on-Year Baby Name Spikes

Name	Gender	Year	Count	% Change
Deneen	F	1964	1572	31340
Aaliyah	F	1994	1412	28140
Mallory	F	1983	658	13060
Tenley	F	2010	650	12900
Tammie	F	1957	584	11580
Jalen	M	1992	583	9617
Cataleya	F	2012	616	7600
Elian	M	2000	542	5922
Deanna	F	1937	1605	5632
Katina	F	1972	2726	5579

While a number of names are likely candidates for being driven by pop-culture influence, the name Aaliyah has the 2nd largest year-on-year spike, with a percentage change of 28 140 in 1994. As the following search indicates, the pop singer Aaliyah’s first entry into the Billboard top 100 chart was also in 1994.

I create the following graph in order to visualise the spike in popularity with and the rise of Aaliyah on the Billboard charts.

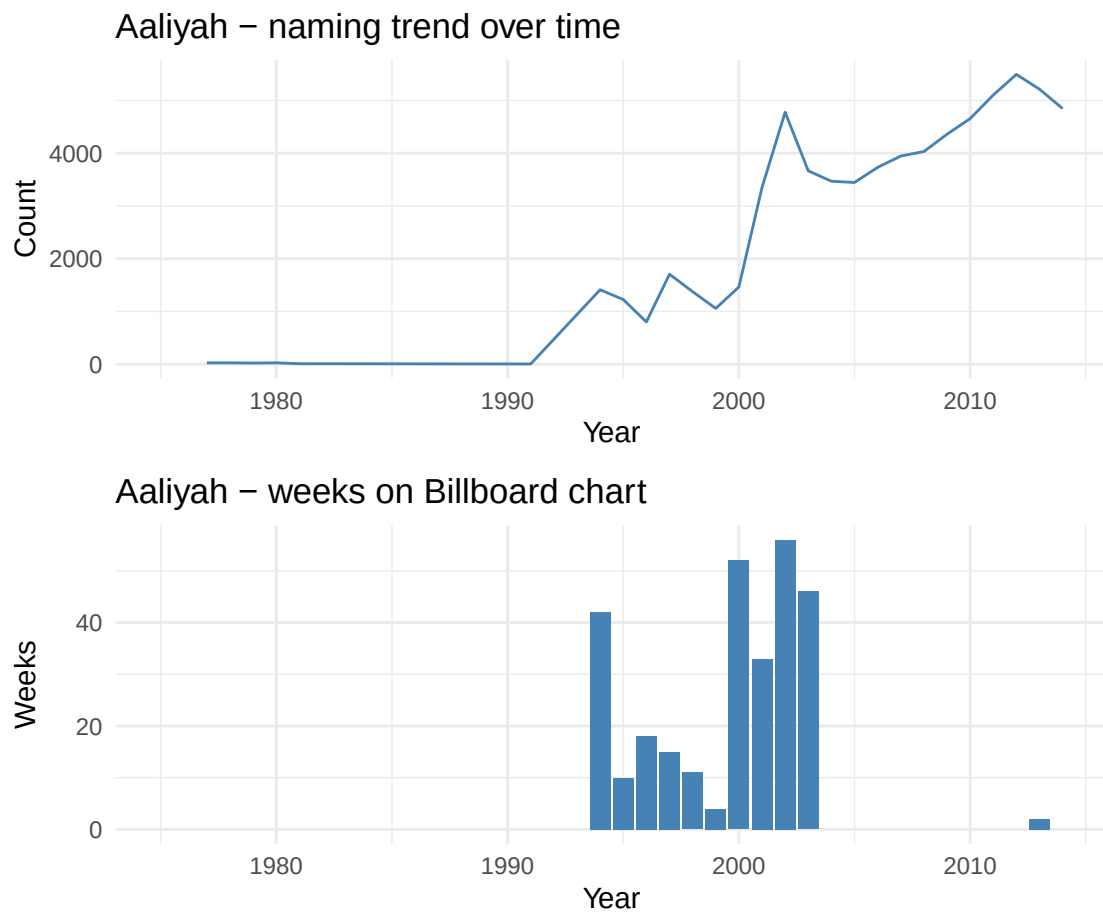


Figure 2.1: The Name Aaliyah, Over Time.

This indicates that pop-culture, fashionable trends, and celebrity culture can influence naming trends over time.